

L24: Goodness-of-Fit Testing

Simulation Without a Black Box

Outline

Purpose of GoF Tests

Kolmogorov-Smirnov Test

Chi-Squared Test

Q-Q Plot

Sample-Size Effect

Key Takeaways

Why Goodness-of-Fit Tests?

The Core Question

After fitting a distribution to data, we ask:

"Is the discrepancy between model and data small enough to be explained by sampling variability alone?"

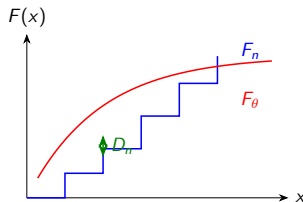
GoF tests answer this formally — but never prove a model is correct. They only **fail to reject** it.

Three complementary tools:

1. Kolmogorov-Smirnov (KS) — CDF-based, exact for continuous
2. Chi-squared (χ^2) — bin-based, approximate
3. Q-Q plot — graphical, sensitive to tails

Warning

Using only one test is risky. A distribution can pass KS but fail visually in the tails.



KS Test: Geometric Interpretation

KS Statistic

$$D_n = \sup_x |F_n(x) - F_\theta(x)|$$

F_n is the empirical CDF; F_θ is the fitted theoretical CDF.

Geometric reading: D_n is the *maximum vertical gap* between the ECDF staircase and the smooth fitted CDF.

Critical values at $\alpha = 0.05$:

n	$D_{n,0.05}$	Reject if $D_n >$
50	0.192	0.192
100	0.136	0.136
500	0.061	0.061

Approximation for large n : $D_{n,0.05} \approx 1.36/\sqrt{n}$.

```
from scipy import stats
```

```
# Fit gamma to data x
```

```
a, loc, sc = stats.gamma.fit(x,
                             floc=0)
```

```
# KS test against fitted CDF
```

```
stat, pval = stats.kstest(
    x,
    "gamma",
    args=(a, loc, sc)
)
```

```
print(f"KS stat = {stat:.4f}")
```

```
print(f"p-value = {pval:.4f}")
```

```
if pval > 0.05:
```

```
    print("Fail to reject alpha=0.05")
```

```
else:
```

```
    print("Reject at alpha=0.05")
```

Warning

`stats.kstest` assumes *known* parameters. After MLE, critical values are anti-conservative. Use Lilliefors correction for Exponential/Normal.

Chi-Squared Test: Equal-Probability Bins

Chi-Squared Statistic

$$\chi^2 = \sum_{j=1}^k \frac{(O_j - E_j)^2}{E_j}$$

O_j = observed count; $E_j = n/k$ under equal-probability binning.

Degrees of freedom:

$$df = k - 1 - p$$

p = number of *estimated* parameters.

Distribution	p	df ($k = 10$ bins)
Exponential	1	8
Gamma	2	7
Lognormal	2	7

Rule: each bin should have $E_j \geq 5$; merge if not.

```
import numpy as np
from scipy import stats

k = 10
# Equal-probability bin edges
quantiles = np.linspace(0, 1, k+1)
edges = stats.gamma.ppf(
    quantiles, a, loc, sc)

# Observed counts
obs, _ = np.histogram(x, bins=edges)
exp = np.full(k, len(x)/k)

# Chi-squared statistic
chi2_stat = ((obs-exp)**2/exp).sum()
df = k - 1 - 2 # gamma: 2 params
pval = 1 - stats.chi2.cdf(
    chi2_stat, df)

print(f"chi2 = {chi2_stat:.3f}")
print(f"df = {df}")
print(f"p = {pval:.4f}")
```

Q-Q Plot: Visualizing Tail Fit

Construction:

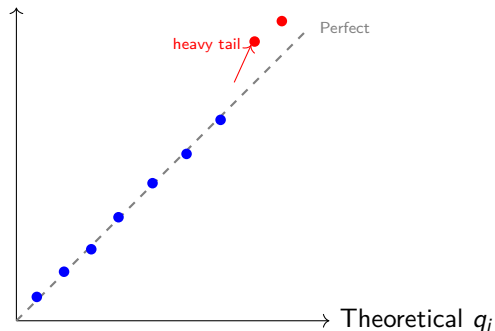
1. Sort data: $x_{(1)} \leq \dots \leq x_{(n)}$
2. Theoretical quantiles: $q_i = F^{-1}\left(\frac{i-0.5}{n}\right)$
3. Plot $(q_i, x_{(i)})$ — perfect fit gives 45° line

Reading deviations:

- **S-shape:** wrong distributional family
- **Upper-tail bow-up:** data heavier-tailed than model
- **Isolated upper point:** outlier, not family mismatch

```
stats.probplot(x, dist="gamma",  
               sparams=(a, loc, sc), plot=plt)  
plt.title("Gamma Q-Q Plot")
```

Sample $x_{(i)}$



The Sample-Size Problem

Box's Maxim

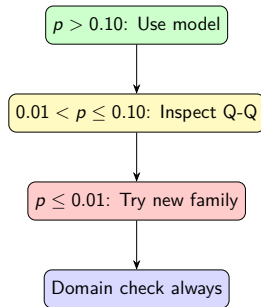
"All models are wrong, but some are useful."

Low n (e.g., 30): KS test has very low power — almost any model passes even if the family is wrong.

High n (e.g., 5000): KS test rejects nearly everything, including the true data-generating process, because MLE estimates are slightly biased and rounding artefacts accumulate.

Decision heuristic:

1. $p > 0.10$: model is adequate
2. $0.01 < p \leq 0.10$: inspect Q-Q; use if tail acceptable
3. $p \leq 0.01$, $n > 200$: usable with domain justification
4. $p \leq 0.01$ and Q-Q shows S-shape: try another family



Warning

Report D_n (effect size) alongside p . Large n guarantees rejection of any parametric model.

Key Takeaways

1. **KS statistic** = max vertical gap between ECDF and fitted CDF;
 $D_{n,0.05} \approx 1.36/\sqrt{n}$ for large n .
2. **Chi-squared test** uses equal-probability bins; $df = k - 1 - p$ where p = estimated parameters. Each bin must satisfy $E_j \geq 5$.
3. **Q-Q plots** are most sensitive to tail mismatch; an S-shape signals a wrong family.
4. **Large samples always reject** — report D_n and Q-Q, not just p .
5. **No test proves correctness** — it only fails to disprove.
6. Use **all three tools** together plus domain knowledge.

Next Lecture

L25 — End-to-end input modeling for the clinic case study.